Seneca Polytechnic - School of Electronics and Mechanical Engineering Technology

TPJ IDEAS **Technical Report** **Quick Guide to Writing An Excellent Report**

TPJ Requirements

Use the comprehensive, technical, report writing skills that you have learned in previous courses (ICO155, COM101, TEC400, etc.) to help you write your own Technical Report this semester.

At a minimum, these 15 major sections and 8 appendices must appear in your final report:

§01. Title Page - include large picture of final project, group members, instructor, course code
§02. Abstract - gets its own page, max. 300 words
§03. Table of Contents (TOC) - containing headings from all sections in document
+ Table of Figures (TOF) - containing captions from all images in document
§04. Introduction - with in-line citations to references
§05. Background Research & Development - with in-line citations to references
§06. System Functional Features & Specifications - technical parameters of whole product
§07. System Design and Layout - SBD and related flowcharts, describes general functionality
§08. Hardware Components & Specifications - useful, fully cited, and with clear bright pics
§09. Theory of Operation - one of the larger sections in the document, provide calculations
§10. Product Operating Instructions - written in clear language, like a recipe
§11. Maintenance Requirements
§12. Future Improvements
§13. Conclusions: Challenges, Successes, Evidential Accuracy - provide proof on how accurate the input readings are and how precisely the outputs will perform
§14. References - MLA format preferred, APA is acceptable too
§15. Appendices
§15A - System Block Diagram - .jpg
§15B - Electrical Diagrams - All Schematics, indicate complete PCB schematic
§15C - PCB Design Diagrams - Any PCB designs, top and bottom, plus real life pics
§15D - Mechanical Diagrams - Any mechanical designs, from multiple angles
§15E - IoT Dashboard - page layout designs, final screenshot images
§15F - Coding Flowchart or Grafcet Diagram
+ Full Project Software Code - must be text, .jpg only if NodeRed or similar
§15G - Bill of Materials - from spreadsheet, insert as clear Table
§15H - Contact Information - name, email, phone number for each team member

Supporting Documentation

If you would like to pursue obtaining your C.E.T. designation from OACETT after graduation, we would recommend first looking at their guidelines for the technical report requirement. The latest version of that document may be found here:
https://www.oacett.org/getmedia/959a2fc6-0d22-4fab-84c4-413a497a3e17/Technology-Report-Guidelines-V3.pdf

Note how the “Structure and Mechanics” items listed on pages 11-12 closely resemble the fifteen sections that are listed above for our TPJ requirements.

Overall Marking Rubric

The following comments are provided to help guide you on producing your best quality work.

#	Criteria for Standard	3	2	1	0
A	**Shell material:** The expected sections (14 + 8 Appendices) are complete, clear, and checked for errors. Research and images are cited in line on each page and in the final Bibliography list of references, in MLA format. Care has been taken to format the overall look of the report: lines justified left, page numbers, and appropriate page breaks. Referencing with footnotes, quotations, and paraphrasing are used correctly.				
B	**Written communications:** All written text is complete, clear, checked for errors, uses a consistent voice and verb tenses, and demonstrates a full collaboration between partners. The report describes accurate and thorough details about the main project topic and its natural progression into associated electronics technologies, using relevant visual components as further illustrations. Meaningful research and connections to recent innovations is provided across multiple sections, demonstrating thoughtful reflection on their influence and larger effects in the technology industry. Content is professionally presented, easy to follow, and matches what was requested for each section.				
C	**Visual documentation:** All imagery is clear, readable, and placed thoughtfully and carefully onto each page with appropriate citations and captions. Images created by the team members use appropriate CAD computer software and are not hand-drawn. Tables created by the team are appropriately spaced, easy to read, and headings are repeated as needed across multiple pages. Students can differentiate between a system block diagram, schematic, wiring diagram, and flowchart.				
D	**Engagement:** All project parts (MCU, IoT, inputs, outputs) are professionally identified with technical specifications that are referenced, relevant, and readable. An advanced student would be able to attempt to recreate this build themselves with the full set of instructions provided in the report. There is clear evidence of relevancy and research, references, and useful visual images. The report is engaging to read and sets a consistent tone.				
E	**Applied improvements:** Thoughtful reflection and evidence of improved writing and re-structuring of any previously submitted content has taken place in the final submission.				

3 – Excellent; **2** – Satisfactory; **1** – Unsatisfactory; **0** – Poor.